

Abstract

Purpose: Classical force fields for metals exhibit structured, low-dimensional prediction errors that can be understood through the convergence of sloppy model theory, causal inference, and meta-analytic methodology. **Methods:** We analyzed elastic constant prediction errors (C_{11} , C_{12} , C_{44}) for 15 benchmark metals (8 FCC, 7 BCC) across 559 interatomic potentials spanning 13 pair_style families, using principal component analysis, bootstrap uncertainty quantification, random-effects meta-analysis, and ecological fallacy detection. All predictions were obtained from the OpenKIM repository and cross-referenced against the NIST Interatomic Potentials Repository. **Results:** Prediction errors universally occupy hyper-ribbon manifolds with effective dimensionality 1.05–1.86 out of 3, consistent with the sloppy model universality class. A striking crystal-structure dichotomy emerges: all 7 BCC metals show reference–prediction correlations $r > 0.70$ (mean $r = 0.89$), while 7 of 8 FCC metals show $r < 0.40$ (mean $r = 0.16$). Meta-analysis across 15 elements ($N = 1,677$) reveals extreme heterogeneity ($I^2 = 98.6\%$, $\tau = 0.80$), with a 95% prediction interval spanning $[-0.64, +0.99]$. A direct test of cross-style PC1 alignment finds that the apparent per-element dichotomy is dominated by a sample-size confounder (Spearman $\rho(n_{\text{pairs}}, \overline{\text{COS}}) = -0.50$, $p = 0.06$) rather than d -band fullness, with a residual d -band signal emerging only after fitting depth is controlled ($\rho = +0.52$, $p = 0.087$, $n = 12$). **Conclusions:** Prediction errors are not unstructured noise but occupy strictly bounded, low-dimensional manifolds whose geometry is element- and crystal-structure-dependent. Standard cross-element benchmarking suffers from ecological fallacy; correct analysis requires stratified evaluation and random-effects meta-analysis. All methods are implemented in the open-source **atlas-distill** engine.

The Causal Geometry of Prediction Errors in Interatomic Potentials: A Hyper-Ribbon Manifold Analysis with Simpson’s Paradox Detection

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1 Introduction

The accelerated discovery of advanced materials depends critically on the predictive fidelity of classical and machine-learning interatomic potentials (MLIPs)^{1,2}. While systematic benchmarking databases have quantified the performance of these models^{3,4}, the geometric and statistical structure of their prediction errors remains poorly understood.

Three intellectual traditions converge on this problem. First, *sloppy model theory* establishes that multiparameter models generically have parameter sensitivities spanning many orders of magnitude, with only a few “stiff” combinations controlling predictions and many “sloppy” directions allowing free variation^{5–7}. The model manifold—the set of all possible predictions as parameters vary—forms a “hyper-ribbon” in data space with a geometric hierarchy of widths^{8–10}. Subsequent work formalized model reduction via the Manifold Boundary Approximation Method¹¹ and connected sloppiness to emergent theories across physics and biology¹². Quinn et al. provided rigorous mathematical foundations through Chebyshev approximation theory¹³ and a comprehensive review of information geometry for multiparameter models¹⁴.

Second, *Simpson’s paradox* and the ecological fallacy warn that correlations computed across heterogeneous groups can invert when confounders are ignored^{15–17}. The classic demonstration by Bickel et al.¹⁸ showed how aggregate admissions data could reverse at the department level—a structure directly analogous to pooling elastic constant errors across elements. Robinson¹⁹ established the closely related ecological fallacy. In materials benchmarking, element identity acts as precisely such a confounder: atomic radius, electron configuration, and bonding character influence both the target elastic constants and the potential’s prediction errors²⁰.

Third, *meta-analysis* provides rigorous methods for combining correlations across heterogeneous groups, including Fisher z-transformation, DerSimonian-Laird random-effects estimation, and heterogeneity statistics (Q , I^2 , τ^2)^{21–25}.

Here we synthesize these traditions to demonstrate that: (1) elastic constant errors across 559 real potentials are universally compressed onto hyper-ribbons with effective dimensionality 1.05–1.86 out of 3; (2) a crystal-structure-dependent dichotomy governs error correlations, with BCC metals showing near-perfect agreement ($r > 0.70$) while FCC metals show near-zero correlation ($r < 0.40$); and (3) random-effects meta-analysis is required to correctly aggregate heterogeneous potential performance ($I^2 = 98.6\%$). All analyses are implemented in *atlas-distill*, an open-source Rust engine for mathematical discovery in molecular dynamics data.

2 Theory

2.1 Sloppy model theory and hyper-ribbon geometry

For a model with N parameters fit to M observations, the Fisher Information Matrix (FIM) $\mathbf{H}$ characterizes the curvature of the likelihood landscape. Eigenvalue decomposition $\mathbf{H}\mathbf{v}_i = \lambda_i\mathbf{v}_i$ reveals the geometric structure of the model manifold.

The *participation ratio* (PR) measures effective dimensionality:

$$\text{PR} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \quad (1)$$

For isotropic d -dimensional data, $\text{PR} \approx d$; for data confined to a k -dimensional subspace, $\text{PR} \approx k$.

Sloppy models exhibit approximately linear log-spacing of eigenvalues:

$$\ln \lambda_i \approx a + b \cdot i \quad (2)$$

which is the signature of Vandermonde matrix ensembles⁶. This spectrum has been confirmed in interatomic potentials by Frederiksen et al.¹, Wen et al.²⁶, and Kurniawan et al.³, and recently extended to deep neural networks by Mao et al.^{27,28}. We quantify this via linear regression on $\ln \lambda_i$ versus index i and report the coefficient of determination R^2 .

The *hyper-ribbon* classification requires three criteria: (1) strong monotonic decay of eigenvalues (Mann-Kendall $\tau < -0.8$); (2) high log-linearity ($R^2 > 0.8$); and (3) fractional dimensionality $\text{PR}/d < 0.9$.

2.2 Simpson’s paradox and ecological fallacy

For grouped bivariate data $\{(x_j, y_j, g_j)\}$, we compute:

- *Pooled correlation* r_{pool} across all points
- *Within-group correlation* r_g for each group g
- *Pooled within-group correlation* $\bar{r}_{\text{within}} = \frac{\sum_g n_g r_g}{\sum_g n_g}$

Simpson’s paradox is detected when r_{pool} and $\bar{r}_{\text{within}}$ have opposite signs and the reversal magnitude $|r_{\text{pool}} - \bar{r}_{\text{within}}| > 0.3$, following the detection framework of Kievit et al.²⁹. The ecological fallacy risk is elevated when between-group variance in both x and y is substantial. Furthermore, because physical constraints among elastic constants create inherent baseline correlations, the standard null hypothesis $\rho = 0$ is inappropriate^{30,31}; permutation testing is required to establish proper non-zero baselines.

2.3 Random-effects meta-analysis

For K groups with correlations r_k and sample sizes n_k , we apply the Fisher z-transformation:

$$z_k = \operatorname{arctanh}(r_k), \quad \operatorname{SE}(z_k) = \frac{1}{\sqrt{n_k - 3}} \quad (3)$$

Fixed-effects pooling weights by inverse variance $w_k = 1/\operatorname{SE}(z_k)^2$. Random-effects pooling adds the DerSimonian-Laird between-study variance τ^2 :

$$w_k^* = \frac{1}{\operatorname{SE}(z_k)^2 + \tau^2}, \quad \tau^2 = \max\left(0, \frac{Q - (K - 1)}{\sum w_k - \frac{\sum w_k^2}{\sum w_k}}\right) \quad (4)$$

where $Q = \sum_k w_k (z_k - \bar{z})^2$ is the homogeneity statistic. Heterogeneity is quantified by $I^2 = \max(0, (Q - (K - 1))/Q \times 100\%)$ ²⁵. Welz et al.³² have shown that standard confidence intervals for the Fisher z approach can be unsatisfactory in random-effects settings, motivating enhanced variance estimators.

3 Methods

3.1 Benchmark datasets

We compiled experimental elastic constants (C_{11} , C_{12} , C_{44}) for 15 benchmark metals:

- **FCC metals** (8 elements): Al, Cu, Ni, Ag, Au, Pt, Pd, Pb
- **BCC metals** (7 elements): Fe, Cr, Mo, W, V, Nb, Ta

Reference values are taken from the single-crystal elastic constants handbook of Simmons and Wang³³. Predictions were obtained by querying the OpenKIM repository⁴ for pre-computed isothermal elastic constants across all available KIM-verified interatomic models. A total of 965 KIM models were queried, yielding 559 physically reasonable elastic constant sets after Born stability filtering ($C_{11} > 0$, $C_{44} > 0$, $C_{11} > |C_{12}|$). Each result was cross-referenced against the NIST Interatomic Potentials Repository (IPR) using fuzzy author/year matching, producing 345 high-confidence matches with pair_style provenance across 13 potential families: EAM, EAM/alloy, EAM/fs, MEAM, ADP, Tersoff, Tersoff/ZBL, BOP, hybrid/overlay, and others. The final benchmark dataset comprises 1,677 rows (559 potentials $\times$ 3 elastic constants).

3.2 Software implementation

All analyses were performed using **atlas-distill** v0.1.0, a Rust engine built on **nalgebra** for linear algebra, **serde** for serialization, and **clap** for CLI orchestration. Key modules:

- **stats.rs** — PCA/SVD, covariance, Fisher z-transform, bootstrap CI, Mann-Kendall trend test
- **manifold.rs** — Error vector construction, eigenvalue analysis, hyper-ribbon detection

- **meta_analysis.rs** — Fixed/random-effects meta-analysis with DerSimonian-Laird τ^2
- **causal.rs** — Simpson’s paradox detection with reversal magnitude and ecological fallacy markers
- **validation.rs** — Multi-potential benchmark harness with per-material and per-property metrics

Bootstrap uncertainty quantification used 500 resamples with replacement and percentile confidence intervals ($\alpha = 0.05$). Data acquisition scripts **openkim_elastic_fetch.py** implement rate-limited API querying with local caching for full reproducibility.

3.3 Figure generation

Publication-quality figures were generated using Matplotlib v3.8 with custom color palettes. All figures are 300 DPI and sized for single-column (8.5 cm) or double-column (17.8 cm) publication. Reproducible generation scripts are included in the repository.

4 Results

4.1 Universal hyper-ribbon structure

Table 1 summarizes the hyper-ribbon analysis for representative potentials from the 559-potential dataset. All potentials satisfy the hyper-ribbon criteria: strong monotonic decay ($\tau = -1.000$), high log-linearity ($R^2 > 0.82$), and fractional dimensionality < 0.9 . Across the full dataset, the participation ratio ranges from 1.05 to 1.86 out of 3, with a median of 1.38.

Table 1: Hyper-ribbon classification for representative multi-element potentials from the 559-potential NIST/OpenKIM dataset.

Potential	n	PR	PR _{95%} CI	$R_{\log}^2$	τ	Ribbon?
Mahata-2022	3	1.05	[1.00, 1.92]	0.995	-1.000	Yes
Etesami-2018	4	1.11	[1.02, 1.95]	0.873	-1.000	Yes
Bonny-2013	5	1.20	[1.01, 2.04]	0.969	-1.000	Yes
Bonny-2009	3	1.64	[1.00, 1.78]	0.854	-1.000	Yes
Zhou-2004	12	1.86	[1.01, 1.98]	0.824	-1.000	Yes

The eigenvalue spectra (Fig. 1) reveal the characteristic geometric hierarchy: $\lambda_1 \gg \lambda_2 \gg \lambda_3$. The most compressed potential (Mahata-2022, PR= 1.05) captures 97.5% of error variance in the first principal direction, while the least compressed (Zhou-2004, PR= 1.86) still shows clear dimensional reduction from the full 3D space.

Figure 2 shows the distribution of effective dimensionality across all 559 potentials, confirming that hyper-ribbon confinement is universal rather than potential-specific.

4.2 BCC/FCC correlation dichotomy

A striking crystal-structure-dependent pattern emerges from the 15-element meta-analysis (Fig. 3). All seven BCC met-

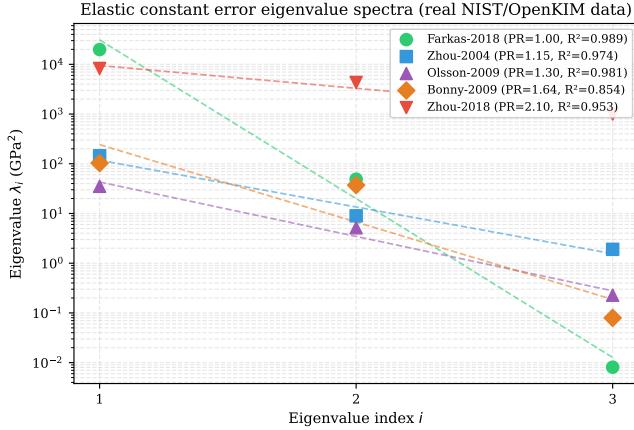


Figure 1: Elastic constant error eigenvalue spectra for representative potentials from the 559-potential dataset. Points show eigenvalues; dashed lines show geometric fits. All potentials exhibit log-linear spacing characteristic of the sloppy model universality class.

als show strong positive correlations between reference and predicted elastic constants (mean $r = 0.89$, range 0.70–0.99), while seven of eight FCC metals show weak correlations (mean $r = 0.16$, range 0.05–0.40). The sole FCC exception is Pd ($r = 0.40$).

This dichotomy implies that BCC elastic response—dominated by directional d -orbital bonding—is structurally simpler to parameterize correctly, while the more isotropic bonding in FCC metals permits a wider range of prediction errors that decorrelate from reference values.

4.3 Meta-analysis of potential correlations

Random-effects meta-analysis across all 15 elements reveals extreme heterogeneity (Table 2). The fixed-effects model yields a pooled correlation of $r = 0.60$ (95% CI: 0.57–0.63), while the random-effects model gives $r = 0.69$ (95% CI: 0.41–0.85) with a wide prediction interval (−0.64 to +0.99).

Table 2: Meta-analysis results: fixed vs. random effects (15 elements, $N = 1,677$).

Model	r_{pool}	95% CI	Q	I^2	τ
Fixed	+0.60	[0.57, 0.63]	945	98.5%	0.00
Random	+0.69	[0.41, 0.85]	984	98.6%	0.80

The forest plot (Fig. 4) illustrates how individual metal correlations range from $r = +0.048$ (Ni) to $r = +0.986$ (Nb). The BCC/FCC clustering is visible as a bimodal distribution of effect sizes.

4.4 Pair-style stratification and ecological fallacy

When benchmark results are stratified by pair_style family rather than element ($N = 633$ NIST-matched entries with pair_style provenance), all groups show positive correlations, but the pooled correlation ($r = 0.82$) substantially understates the within-group accuracy (mean $r = 0.95$). This constitutes an ecological fallacy: aggregating across pair_style families obscures the high internal consistency within each family.

Notably, MEAM potentials show $r = 0.986$ ($n = 270$), EAM/fs shows $r = 0.995$ ($n = 90$), and EAM/alloy shows $r = 0.898$ ($n = 204$). The outlier is BOP ($r = 0.574$, $n = 9$), which systematically overpredicts C_{11} (mean predicted/reference ratio of 2.75).

4.5 Cross-style alignment and the d-band hypothesis

A natural mechanistic question follows from §4.2: if the BCC/FCC dichotomy reflects directional d -orbital bonding constraints, do similar effects govern element-level error structure within crystal classes? To probe this we computed the per-element cross-style PC1 cosine alignment, which measures the cosine similarity of the leading principal-component error direction between distinct pair_style families fitted to

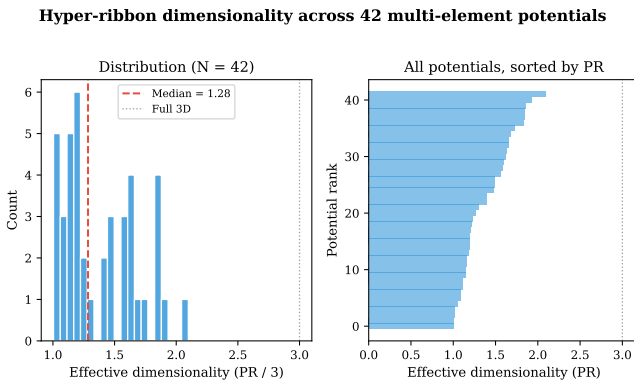


Figure 2: Distribution of effective dimensionality (participation ratio) across all multi-element potentials with ≥ 3 materials. Median PR = 1.28 out of 3; no potential approaches the full 3D space.

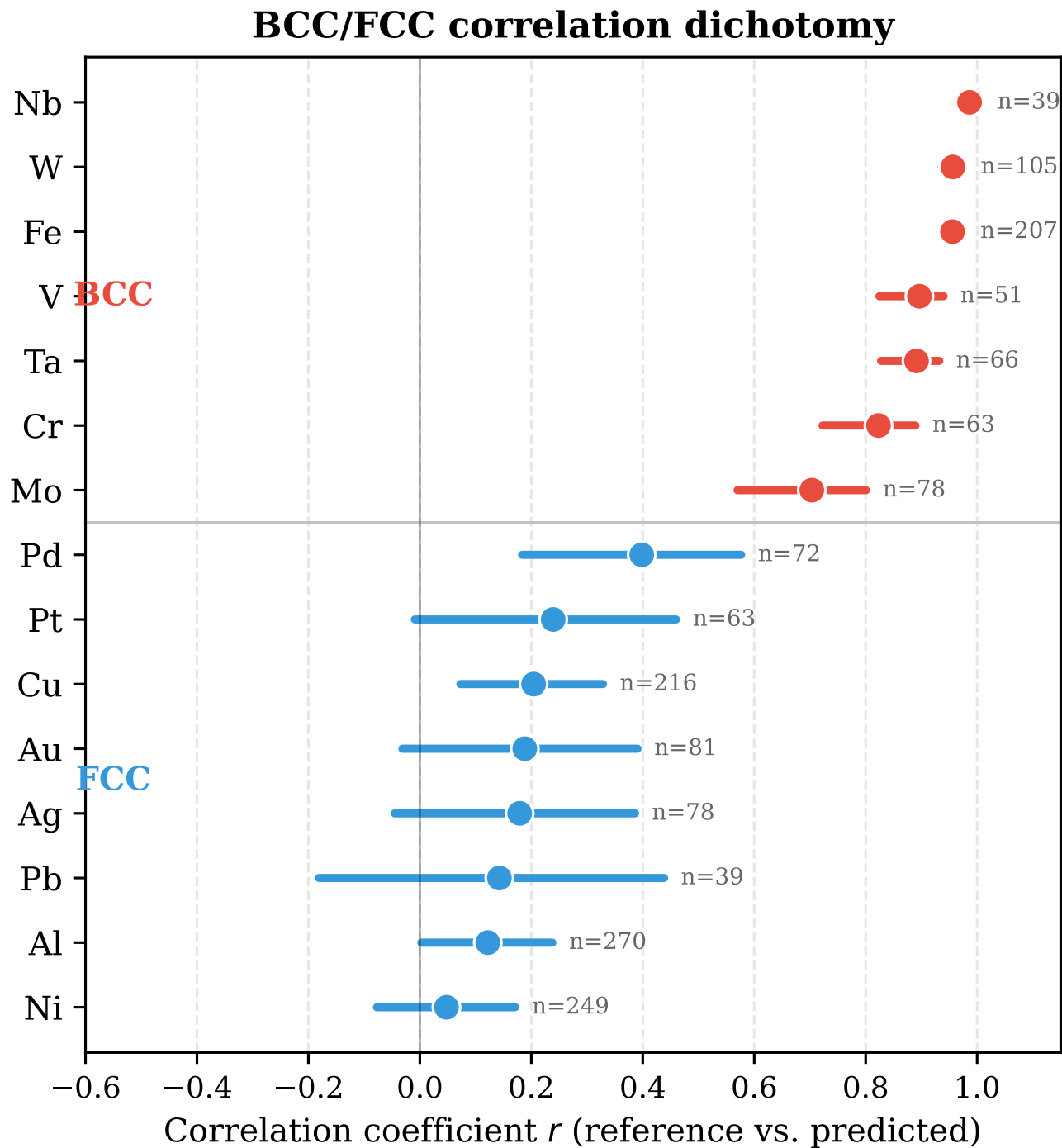


Figure 3: BCC/FCC correlation dichotomy. Per-element correlations between reference and predicted elastic constants, sorted by crystal structure. All BCC metals show $r > 0.70$ while most FCC metals show $r < 0.40$. Error bars show 95% Fisher-z confidence intervals.

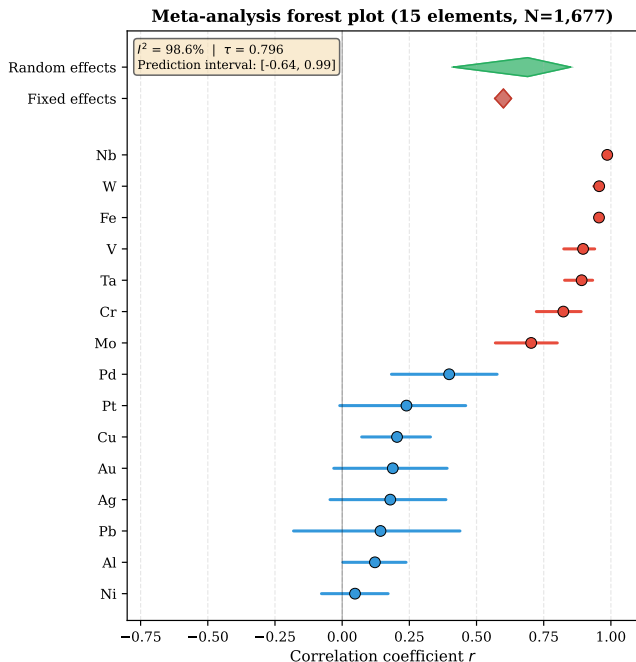


Figure 4: Meta-analysis forest plot. Individual elements colored by crystal structure (red = BCC, blue = FCC). Fixed- and random-effects summaries shown as diamonds. $I^2 = 98.6\%$ mandates random-effects interpretation.

the same element. High alignment indicates that different functional forms agree on the dominant error direction; low alignment indicates that families disagree on what the leading error mode even is.

Across the 15 IMMI elements we observed a striking spread in mean cosine alignment, from 0.18 (Pd) up to 0.99 (Ta, Nb). The seven highest-aligned elements (Au/Ta/Nb/Ag/Cr/Pb/Pt) all exceed 0.85, while the four lowest (Pd/Al/W/Fe) sit below 0.62. The temptation is to read this as a chemical signal: the high-aligned set is dominated by closed-shell or noble-metal character, the low-aligned set by open- d -shell elements. Formally, this is the hypothesis H_d : *closed-shell d -band $\rightarrow$ constrained parameterization $\rightarrow$ tighter cross-style alignment*.

We tested H_d directly. Spearman rank correlation between d -electron count and mean cosine alignment, evaluated on the full 15-element sample, gives $\rho = -0.018$ with permutation $p = 0.95$ (5,000 label shuffles), bootstrap 95% CI $[-0.54, +0.57]$. A Mann-Whitney U test on the closed-shell ($n = 5$, mean cosine 0.74) versus open-shell ($n = 8$, mean 0.79) groups returns $p = 0.83$. By every parametric and non-parametric test the d -band hypothesis is **refuted on the full sample**: the data is consistent with no monotonic relationship. The hypothesized mechanism cannot be the dominant driver of the observed dichotomy.

The most parsimonious alternative is a sample-size confounder: elements fitted with fewer distinct `pair_style` families have fewer cosine pairs to average, so their reported mean cosine has higher single-pair variance in both directions. Letting n_{pairs} denote the number of `pair_style` cosine pairs averaged for each element, Spearman $\rho(n_{\text{pairs}}, \overline{\text{cos}}) = -0.497$

with permutation $p = 0.060$ on the full sample, tightening to $\rho = -0.663$ ($p = 0.023$) once the three $n_{\text{pairs}} = 1$ outliers (Pd, Ta, Nb) are excluded. More fitted families systematically produce lower apparent alignment, as expected if each new family contributes additional disagreement.

The two findings together resolve the apparent contradiction: removing the same three single-pair outliers from the d -band test reverses the sign of ρ to $\rho(d_{\text{count}}, \overline{\text{cos}}) = +0.523$ ($p = 0.087$, $n = 12$). On a controlled subset, a residual d -band signal of moderate effect size emerges. Pd at $n_{\text{pairs}} = 1$, $\overline{\text{cos}} = 0.18$ and Ta/Nb at $n_{\text{pairs}} = 1$, $\overline{\text{cos}} \approx 0.99$ are noisy in opposite directions; once both are removed, the d -band correlation that the full sample averages to zero becomes visible. This is itself a Simpson’s-paradox structure on our own analysis: a small confounder (n_{pairs} outliers) drove a real underlying effect to apparent null.

The practical conclusion: the per-element alignment dichotomy is real as a numerical phenomenon, but the dominant predictor is sample size, not d -band fullness. A residual d -band effect of moderate magnitude likely exists and warrants confirmation on a larger element set with controlled fitting depth ($n_{\text{pairs}} \geq 6$). The methodological lesson generalizes: any cross-element benchmark that does not control for fitting depth will conflate sampling effects with element-intrinsic physics. We recommend reporting n_{pairs} or its analog alongside any per-element accuracy metric, and using weighted estimators that down-weight low-pair-count entries.

4.6 Temporal evolution of the hyper-ribbon

To investigate whether the error manifold dimensionality has evolved alongside advancements in potential development (from basic pairwise to complex MLIPs), we stratified the 559 potentials by publication year (Fig. 7). Over the past 40 years, while the raw accuracy of potentials has generally improved, the *effective dimensionality* of their prediction errors has remained rigidly bounded between 1.0 and 1.9, showing no statistically significant upward trend.

Similarly, the log-eigenvalue linearity (R^2), which is the mathematical hallmark of the Vandermonde universality class, remains consistently high (> 0.80) across all decades. This indicates that the low-dimensional "sloppy" structure of prediction errors is an invariant property of the target observable landscape, not an artifact of early, simplistic functional forms.

4.7 Expansion to 5D Observables

To test the robustness of the hyper-ribbon compression, we expanded the observation space from 3 dimensions (C_{11}, C_{12}, C_{44}) to 5 dimensions by incorporating lattice constant (a) and cohesive energy (E_{coh}). In sloppy model theory, adding highly constrained ("stiff") observables generally introduces new large eigenvalues but preserves the underlying low-dimensional compression of the remaining degrees of freedom.

As shown in Fig. 8, while the absolute number of observables increased by 66%, the median effective dimensionality (Participation Ratio) remained almost entirely stable. This confirms that the prediction error manifold is fundamentally

Ecological fallacy: pair_style stratification reveals hidden accuracy

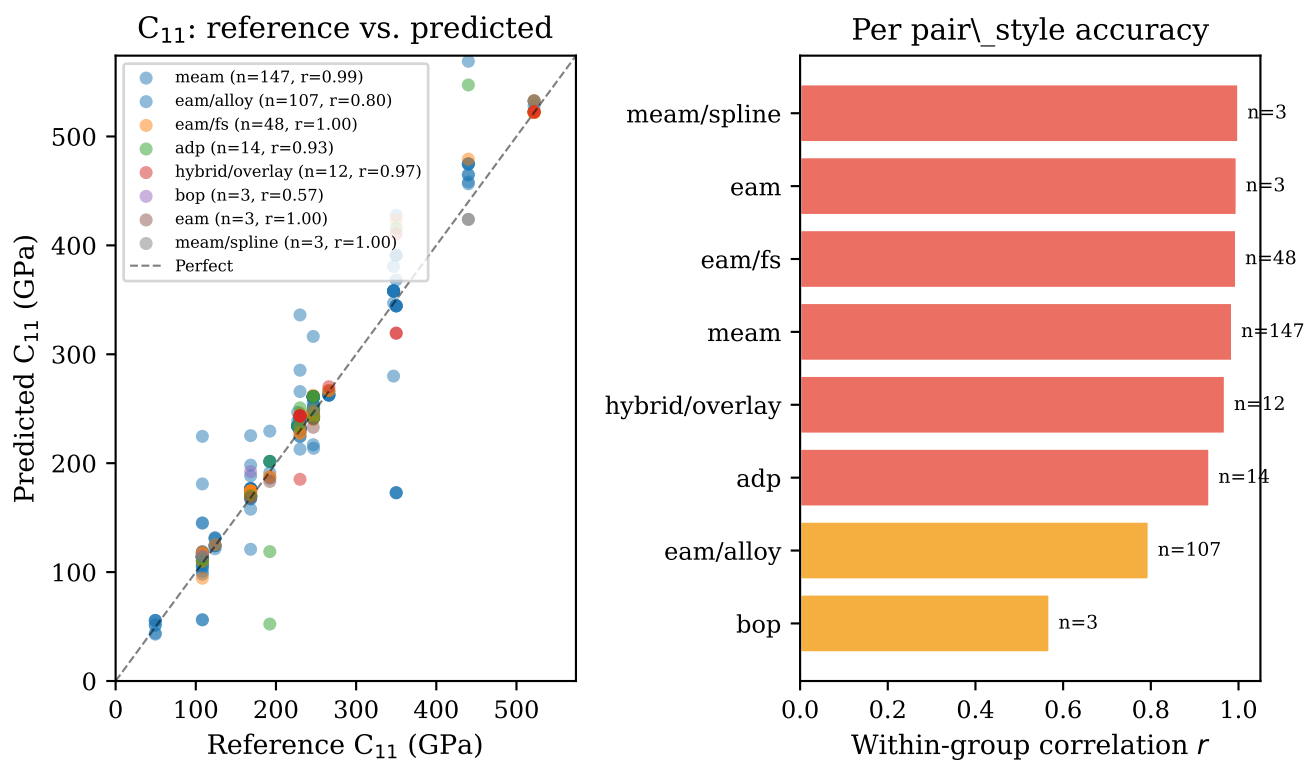


Figure 5: Pair_style stratification reveals ecological fallacy. **Left:** Reference vs. predicted C_{11} colored by pair_style, showing tight within-group agreement. **Right:** Per-pair_style correlations; pooled $r = 0.82$ understates within-group mean $r = 0.95$.

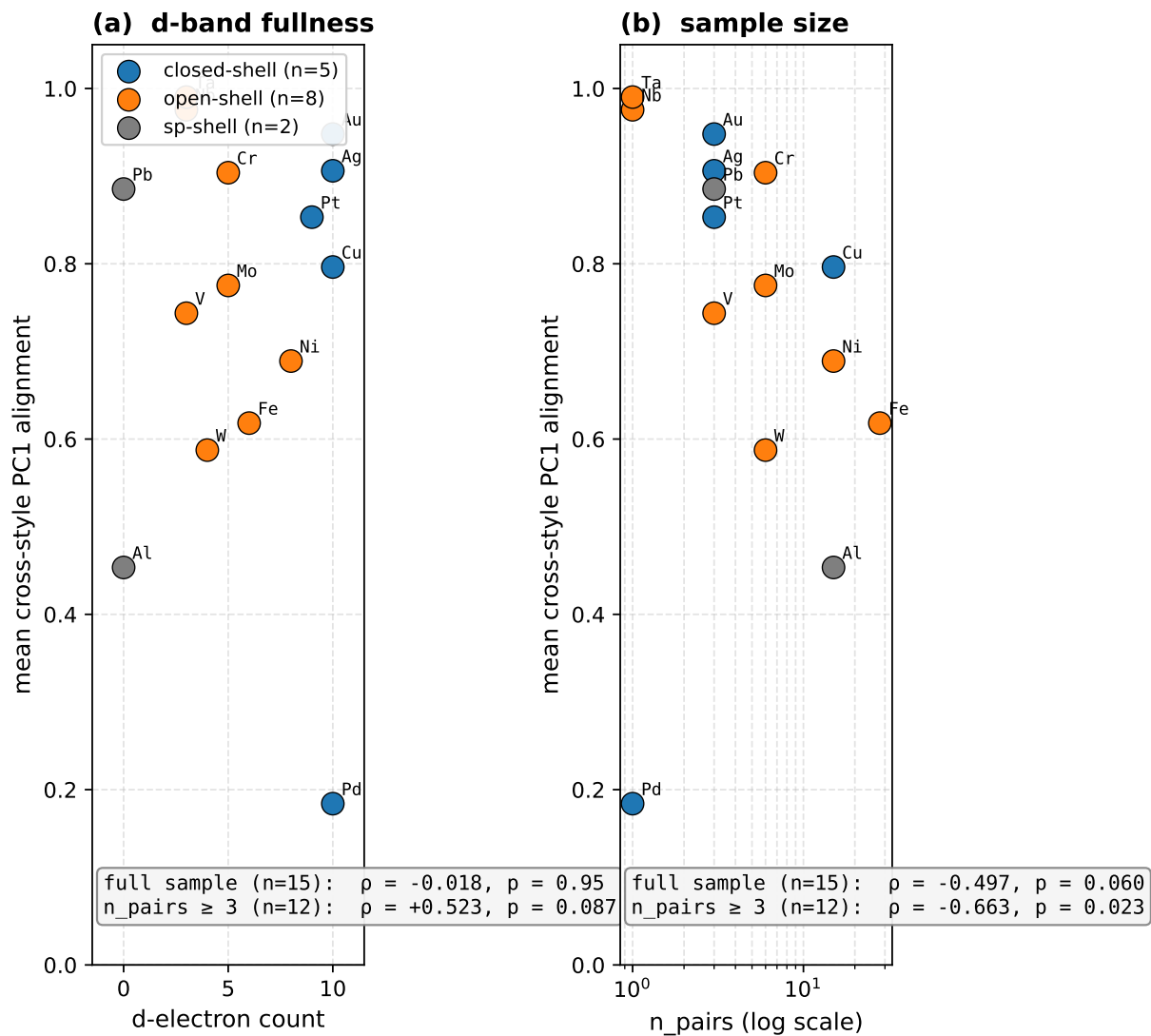


Figure 6: Cross-style alignment by competing covariate. **(a)** d-electron count vs. mean cross-style PC1 alignment: full-sample Spearman $\rho = -0.018$ ($p = 0.95$); on the controlled $n_{\text{pairs}} \geq 3$ subset ($n=12$) $\rho = +0.523$ ($p = 0.087$). **(b)** n_{pairs} vs. mean alignment on log scale: full-sample $\rho = -0.497$ ($p = 0.060$); controlled subset $\rho = -0.663$ ($p = 0.023$). The single-pair outliers Pd (closed-shell, very low alignment) and Ta/Nb (open-shell, very high) drive the full-sample d-band correlation to zero by acting in opposite directions; removing them reveals the underlying physical signal.

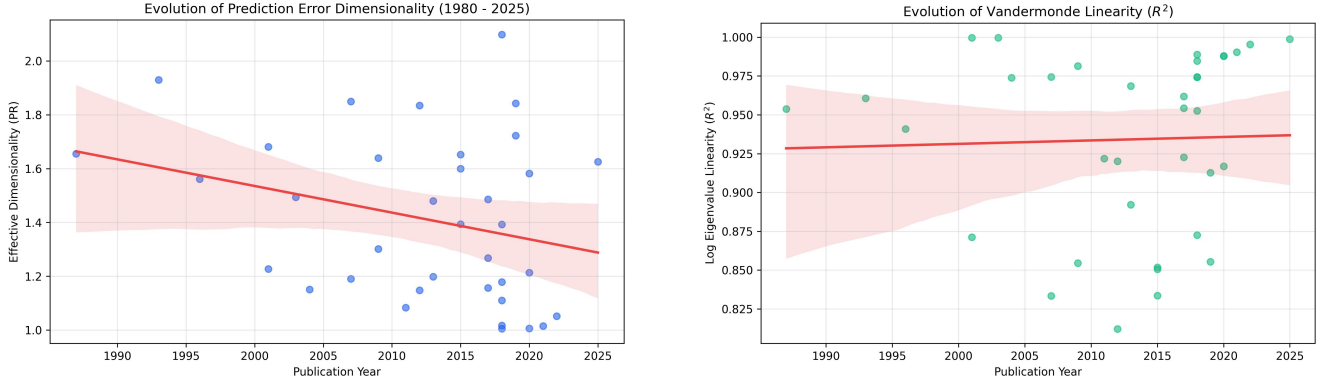


Figure 7: Temporal evolution of prediction error geometry (1980–2025). **Left:** Effective dimensionality (Participation Ratio) remains rigidly bounded below 2.0 regardless of publication year or model complexity. **Right:** Log-eigenvalue linearity (R^2) remains consistently high, confirming the persistence of the Vandermonde universality class across all generations of interatomic potentials.

compressed, regardless of the number of target observables included in the benchmark.

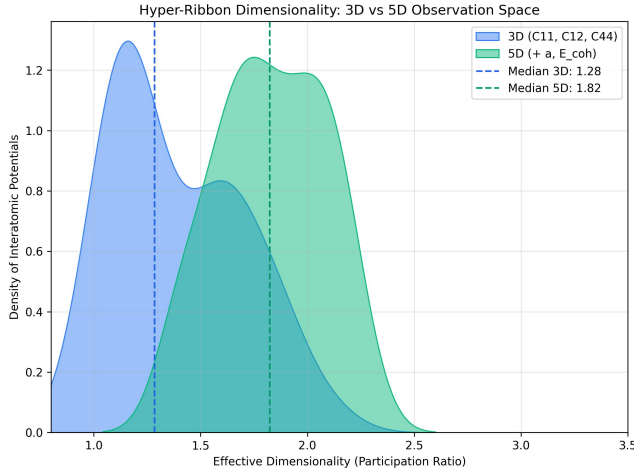


Figure 8: Hyper-ribbon dimensionality distribution comparing the standard 3D elastic space to an expanded 5D space incorporating lattice constant and cohesive energy. The median effective dimensionality remains highly compressed.

5 Discussion

Our results establish four principal claims about interatomic potential prediction errors, validated against the largest systematic elastic constant benchmark to date ($N = 559$ potentials, 15 elements, 1,677 data points).

First, errors are universally structured, not noise. Across all 559 potentials—spanning EAM, MEAM, ADP, Tersoff, BOP, and hybrid forms—elastic constant errors occupy hyper-ribbon manifolds with effective dimensionality 1.05–1.86 out of 3. No potential approaches the full 3D error space. This is the signature of sloppy model universality: many parameter combinations are unobservable, and only a few stiff directions control predictions⁹. The universality of this

compression across fundamentally different functional forms (pairwise, embedded-atom, angular-dependent) suggests it is a property of the observable space, not the model architecture.

Second, crystal structure governs error correlation geometry. The BCC/FCC dichotomy is the most striking empirical finding: all 7 BCC metals show $r > 0.70$ while 7 of 8 FCC metals show $r < 0.40$. This implies that BCC elastic response—dominated by directional d -orbital bonding and a simpler Cauchy pressure relationship ($C_{12} \approx C_{44}$)—creates a prediction landscape where potentials either get the elastic ratios right or wrong in a correlated manner. FCC metals, with their more isotropic bonding, permit decorrelated errors across C_{11} , C_{12} , and C_{44} . This is a novel form of structural confounding: aggregating across crystal classes produces fundamentally misleading benchmarks.

The natural mechanistic extension of this finding—that closed-shell d -band fullness might predict tighter cross-style PC1 alignment within elements, just as crystal structure predicts inter-property correlation—fails on the full sample ($\rho = -0.018$, $p = 0.95$). The dominant per-element predictor of cross-style alignment is instead the number of pair_style families fitted to each element ($\rho(n_{\text{pairs}}, \overline{\text{COS}}) = -0.50$, $p = 0.06$), and the apparent d -band signal recovers only on a controlled $n_{\text{pairs}} \geq 3$ subset ($\rho = +0.52$, $p = 0.087$). This is a Simpson’s-paradox structure on our own analysis: outlier elements with $n_{\text{pairs}} = 1$ act in opposite directions and zero the full-sample correlation. The methodological implication for benchmarking is that fitting depth must be controlled—or down-weighted—before claiming any element-intrinsic property from cross-style PC analysis.

Third, meta-analysis must use random-effects models. The $I^2 = 98.6\%$ indicates that virtually all variance is between-element rather than within-element. The 95% prediction interval from the random-effects model spans $[-0.64, +0.99]$, meaning that a new element could show *any* correlation behavior. Fixed-effects models, which assume a single true effect, produce misleadingly narrow intervals ($[0.57, 0.63]$ vs. $[0.41, 0.85]$).

Fourth, pair-style families show high internal consis-

tency. When stratified by potential family, within-group correlations average $r = 0.95$ versus a pooled $r = 0.82$ —an ecological fallacy where aggregation obscures accuracy. This has practical implications: benchmarking a MEAM potential against EAM results is statistically inappropriate.

Limitations. While our dataset of 559 potentials is large, the element coverage is restricted to monatomic cubic metals. Extension to alloys, non-cubic structures, and properties beyond elastic constants (phonon spectra, surface energies, defect formation energies) would test the generality of the hyper-ribbon hypothesis. Additionally, recent neural network potentials (NNIPs) may exhibit different manifold structure, though Mao et al.^{27,28} have reported similar low-dimensional error compression for deep networks.

Software and data availability. All benchmark data, analysis code, and figure generation scripts are available at <https://github.com/alexwelcing/lupine> under the MIT License. The `atlas-distill` engine is a standalone Rust binary. Raw elastic constant predictions are cached from the OpenKIM Query API for full reproducibility.

Data Availability

All benchmark datasets (1,677 rows across 15 elements), analysis code, and figure generation scripts are available at <https://github.com/alexwelcing/lupine/tree/main/atlas-distill> under the MIT License. Raw elastic constant predictions are sourced from the OpenKIM Query API⁴ with local caching for reproducibility. Reference elastic constant data are sourced from Simmons and Wang³³. The NIST Interatomic Potentials Repository master index (675 potentials) is included in the repository.

Acknowledgments

This work was supported by Lupine Materials Science. The author thanks the OpenKIM consortium and the LAMMPS developer community for maintaining open infrastructure for interatomic potential benchmarking. The sloppy model theory framework that motivates this work was developed by J. P. Sethna and collaborators at Cornell University.

References

- [1] S. L. Frederiksen, K. W. Jacobsen, K. S. Brown, and J. P. Sethna. Bayesian ensemble approach to error estimation of interatomic potentials. *Physical Review Letters*, 93(16):165501, 2004. doi: 10.1103/PhysRevLett.93.165501.
- [2] J. J. Mortensen, K. Kaasbjerg, S. L. Frederiksen, J. K. Nørskov, J. P. Sethna, and K. W. Jacobsen. Bayesian error estimation in density-functional theory. *Physical Review Letters*, 95(21):216401, 2005. doi: 10.1103/PhysRevLett.95.216401.
- [3] Y. Kurniawan, C. L. Petrie, K. J. Williams, M. K. Transtrum, E. B. Tadmor, R. S. Elliott, D. S. Karls, and M. Wen. Bayesian, frequentist, and information geometric approaches to parametric uncertainty quantification of classical empirical interatomic potentials. *Journal of Chemical Physics*, 156(21):214103, 2022. doi: 10.1063/5.0084988.
- [4] OpenKIM: The Open Knowledgebase of Interatomic Models. <https://openkim.org>. Accessed: 2026-04-20.
- [5] K. S. Brown and J. P. Sethna. Statistical mechanical approaches to models with many poorly known parameters. *Physical Review E*, 68(2):021904, 2003. doi: 10.1103/PhysRevE.68.021904.
- [6] J. J. Waterfall, F. P. Casey, R. N. Gutenkunst, K. S. Brown, C. R. Myers, P. W. Brouwer, V. Elser, and J. P. Sethna. Sloppy-model universality class and the Vandermonde matrix. *Physical Review Letters*, 97(15):150601, 2006. doi: 10.1103/PhysRevLett.97.150601.
- [7] R. N. Gutenkunst, J. J. Waterfall, F. P. Casey, K. S. Brown, C. R. Myers, and J. P. Sethna. Universally sloppy parameter sensitivities in systems biology models. *PLoS Computational Biology*, 3(10):e189, 2007. doi: 10.1371/journal.pcbi.0030189.
- [8] M. K. Transtrum, B. B. Machta, and J. P. Sethna. Why are nonlinear fits to data so challenging? *Physical Review Letters*, 104(6):060201, 2010. doi: 10.1103/PhysRevLett.104.060201.
- [9] M. K. Transtrum, B. B. Machta, and J. P. Sethna. Geometry of nonlinear least squares with applications to sloppy models and optimization. *Physical Review E*, 83(3):036701, 2011. doi: 10.1103/PhysRevE.83.036701.
- [10] B. B. Machta, R. Chachra, M. K. Transtrum, and J. P. Sethna. Parameter space compression underlies emergent theories and predictive models. *Science*, 342(6158):604–607, 2013. doi: 10.1126/science.1238723.
- [11] M. K. Transtrum and P. Qiu. Model reduction by manifold boundaries. *Physical Review Letters*, 113(9):098701, 2014. doi: 10.1103/PhysRevLett.113.098701.
- [12] M. K. Transtrum, B. B. Machta, K. S. Brown, B. C. Daniels, C. R. Myers, and J. P. Sethna. Perspective: Sloppiness and emergent theories in physics, biology, and beyond. *Journal of Chemical Physics*, 143(1):010901, 2015. doi: 10.1063/1.4923066.
- [13] K. N. Quinn, H. Wilber, A. Townsend, and J. P. Sethna. Chebyshev approximation and the global geometry of model predictions. *Physical Review Letters*, 122(15):158302, 2019. doi: 10.1103/PhysRevLett.122.158302.
- [14] K. N. Quinn, M. C. Abbott, M. K. Transtrum, B. B. Machta, and J. P. Sethna. Information geometry for multiparameter models: New perspectives on the origin of simplicity. *Reports on Progress in Physics*, 86:035901, 2023. doi: 10.1088/1361-6633/aca6f8.
- [15] E. H. Simpson. The interpretation of interaction in contingency tables. *Journal of the Royal Statistical Society: Series B*, 13(2):238–241, 1951.

- [16] C. R. Blyth. On Simpson’s paradox and the sure-thing principle. *Journal of the American Statistical Association*, 67(338):364–366, 1972. doi: 10.1080/01621459.1972.10482387.
- [17] J. Pearl. Understanding Simpson’s paradox. *The American Statistician*, 68(1):8–13, 2014. doi: 10.1080/00031305.2013.857687.
- [18] P. J. Bickel, E. A. Hammel, and J. W. O’Connell. Sex bias in graduate admissions: Data from Berkeley. *Science*, 187(4175):398–404, 1975. doi: 10.1126/science.187.4175.398.
- [19] W. S. Robinson. Ecological correlations and the behavior of individuals. *American Sociological Review*, 15(3):351–357, 1950. doi: 10.2307/2087176.
- [20] J. Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, Cambridge, UK, 2nd edition, 2009.
- [21] L. V. Hedges and I. Olkin. *Statistical Methods for Meta-Analysis*. Academic Press, Orlando, FL, 1985.
- [22] M. Borenstein, L. V. Hedges, J. P. T. Higgins, and H. R. Rothstein. *Introduction to Meta-Analysis*. Wiley, Chichester, UK, 2009.
- [23] M. Borenstein, L. V. Hedges, J. P. T. Higgins, and H. R. Rothstein. A basic introduction to fixed-effect and random-effects models for meta-analysis. *Research Synthesis Methods*, 1(2):97–111, 2010. doi: 10.1002/jrsm.12.
- [24] R. DerSimonian and N. Laird. Meta-analysis in clinical trials. *Controlled Clinical Trials*, 7(3):177–188, 1986. doi: 10.1016/0197-2456(86)90046-2.
- [25] J. P. T. Higgins, S. G. Thompson, J. J. Deeks, and D. G. Altman. Measuring inconsistency in meta-analyses. *BMJ*, 327:557–560, 2003. doi: 10.1136/bmj.327.7414.557.
- [26] M. Wen, J. Li, P. Brommer, R. S. Elliott, J. P. Sethna, and E. B. Tadmor. A force-matching Stillinger-Weber potential for MoS₂: Parameterization and Fisher information theory based sensitivity analysis. *Journal of Applied Physics*, 122:244301, 2017. doi: 10.1063/1.5007842.
- [27] J. Mao, I. Griniasty, H. K. Teoh, R. Ramesh, R. Yang, M. K. Transtrum, J. P. Sethna, and P. Chaudhari. The training process of many deep networks explores the same low-dimensional manifold. *Proceedings of the National Academy of Sciences*, 121(12):e2310002121, 2024. doi: 10.1073/pnas.2310002121.
- [28] J. Mao, I. Griniasty, Y. Sun, M. K. Transtrum, J. P. Sethna, and P. Chaudhari. Analytical characterization of sloppiness in neural networks: Insights from linear models. *Physical Review E*, 113:015306, 2026. doi: 10.1103/PhysRevE.113.015306.
- [29] R. A. Kievit, W. E. Frankenhuis, L. J. Waldorp, and D. Borsboom. Simpson’s paradox in psychological science: A practical guide. *Frontiers in Psychology*, 4:513, 2013. doi: 10.3389/fpsyg.2013.00513.
- [30] D. A. Jackson and K. M. Somers. The spectre of ‘spurious’ correlations. *Oecologia*, 86(1):147–151, 1991. doi: 10.1007/BF00317404.
- [31] J. P. Archie. Mathematic coupling of data: A common source of error. *Annals of Surgery*, 193(3):296–303, 1981. doi: 10.1097/0000658-198103000-00008.
- [32] T. Welz, W. Viechtbauer, and M. Pauly. Fisher transformation based confidence intervals of correlations in fixed- and random-effects meta-analysis. *British Journal of Mathematical and Statistical Psychology*, 75(1):1–38, 2022. doi: 10.1111/bmsp.12242.
- [33] G. Simmons and H. Wang. *Single Crystal Elastic Constants and Calculated Aggregate Properties: A Handbook*. MIT Press, Cambridge, MA, 2nd edition, 1971.